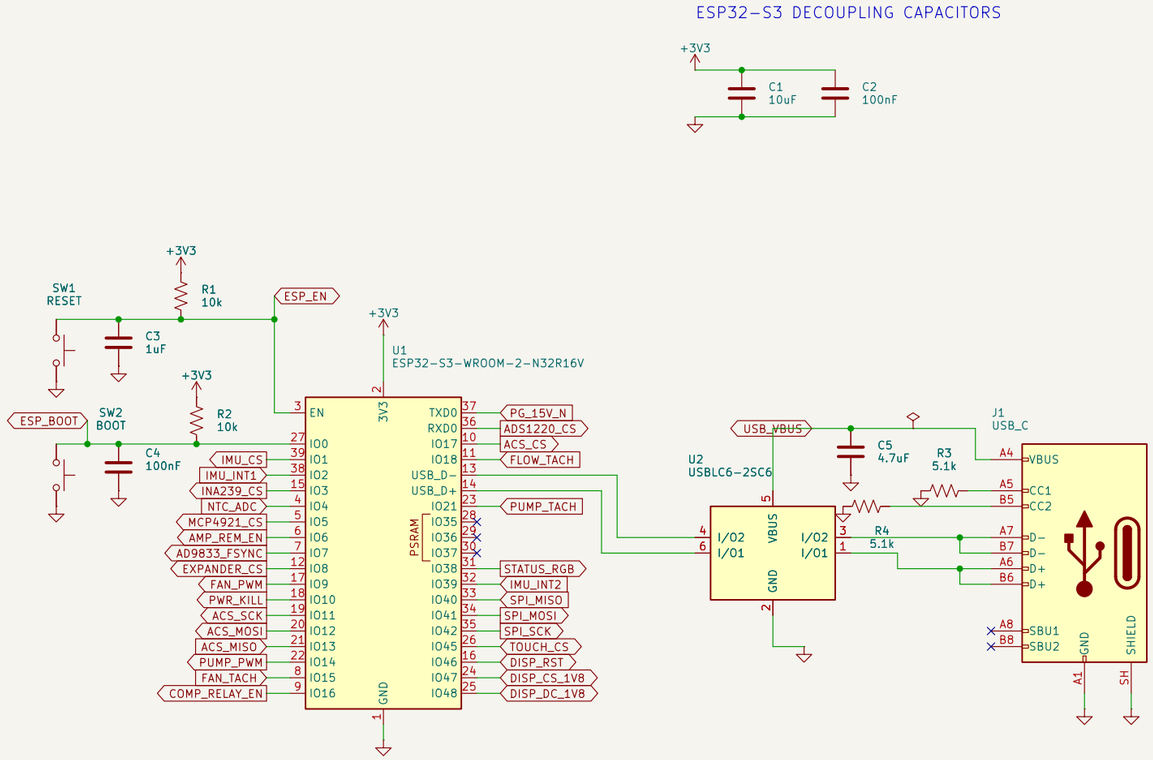
ESP32-S3-WROOM-2-N32R16V: 32 MB octal flash + 16 MB octal PSRAM (GPIO26-37 reserved by module)

GPIO map follows include/config/pin_config.h




Power Input
File: sheets/power_input.kicad_sch

Signal Generator
File: sheets/signal_gen.kicad_sch

Water Cooling
File: sheets/cooling.kicad_sch

Power Button
File: sheets/power_button.kicad_sch

Amp Control
File: sheets/amp_control.kicad_sch

Cold Head Temp
File: sheets/cold_head.kicad_sch

Rails 5V 3V3
File: sheets/rails_logic.kicad_sch

Amp Current
File: sheets/amp_current.kicad_sch

IMU
File: sheets/imu.kicad_sch

Rails +-15V
File: sheets/rails_15v.kicad_sch

Power Good
File: sheets/power_good.kicad_sch

System Current
File: sheets/system_current.kicad_sch

Indicators
File: sheets/indicators.kicad_sch

Compressor Relay
File: sheets/compressor_relay.kicad_sch

GPIO Expander
File: sheets/gpio_expander.kicad_sch

Debug
File: sheets/debug.kicad_sch

Supply Switches

Display

**Cryocooler Controller**

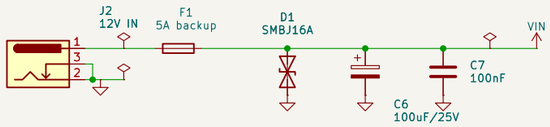
Sheet: /
File: cryocooler.kicad_sch

***Title: Cryocooler Controller***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/19 |

12 V input from a Dell XYK93 (DPS-2000EB class, ~12.2 V, ~160 A available): use the main 12 V rail with PS_ON held low.

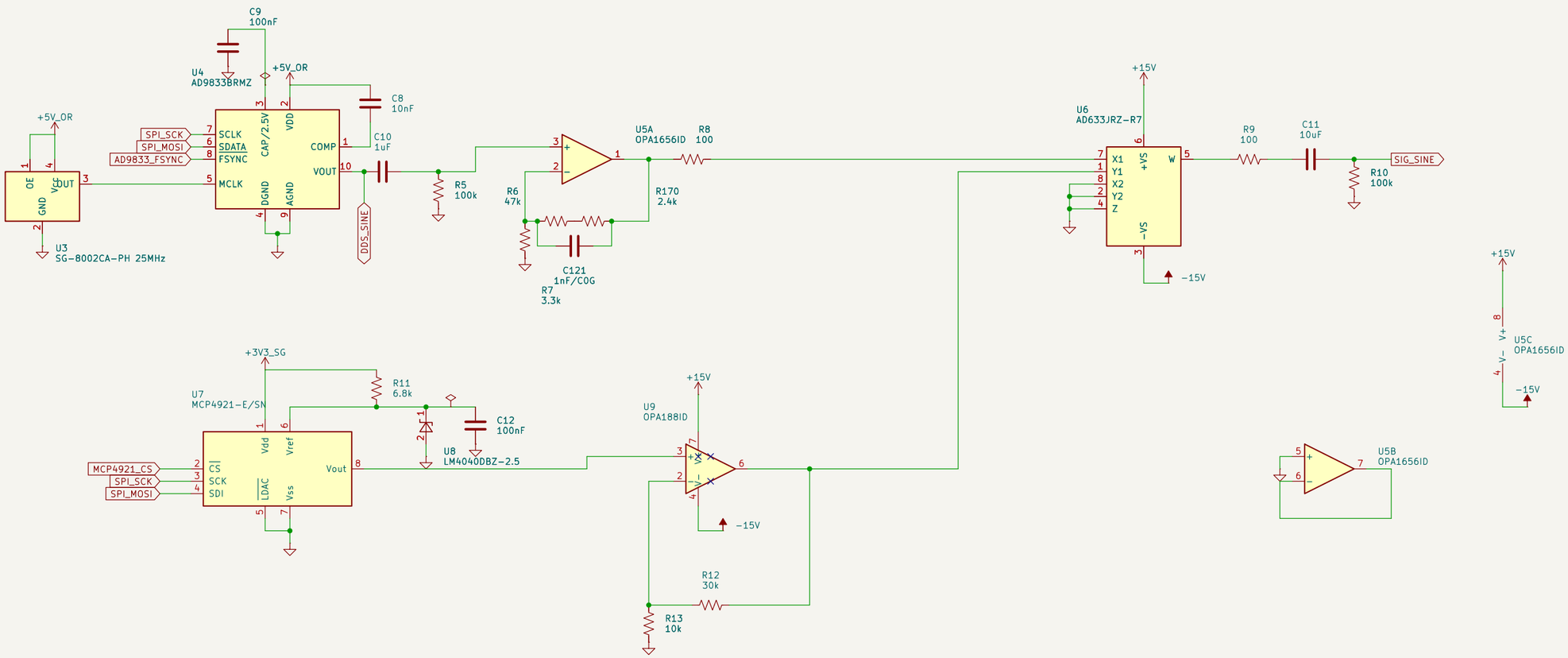
F1 is a backup only: add an inline high-breaking-capacity fuse (7.5-10 A) at the PSU end. Reverse polarity: TVS forward-clamps and blows the fuse.



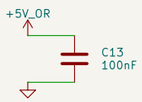

| | | |
|---|---|---|
| **Cryocooler Controller** | | |
| Sheet: /Power Input/<br>File: power_input.kicad_sch | | |
| ***Title: Power Input*** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 2/19 |

AD9833 60 Hz sine -> OPA1656 gain stage -> AD633 multiplier; MCP4921 + OPA188 give a 0–10 V amplitude control.

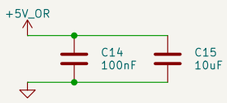
Vctl 0–10 V (2.5 V ref x4) x sine +/−2.3 V pk / 10 = 0..2.3 V pk (4.6 Vpp, 1.6 Vrms) to amplifier input; ZE500.1 RCA input range 0.2–6 V.



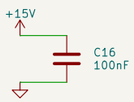

## SG–8002CA DECOUPLING CAPACITORS



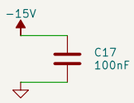

## AD9833 DECOUPLING CAPACITORS



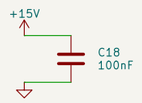

## OPA1656 +15V DECOUPLING CAPACITORS



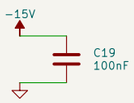

## OPA1656 −15V DECOUPLING CAPACITORS



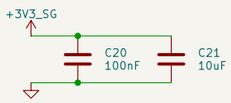

## AD633 +15V DECOUPLING CAPACITORS



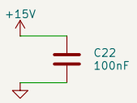

## AD633 −15V DECOUPLING CAPACITORS



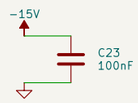

## MCP4921 DECOUPLING CAPACITORS




## OPA188 +15V DECOUPLING CAPACITORS




## OPA188 −15V DECOUPLING CAPACITORS




**Cryocooler Controller**

Sheet: /Signal Generator/
File: signal_gen.kicad_sch

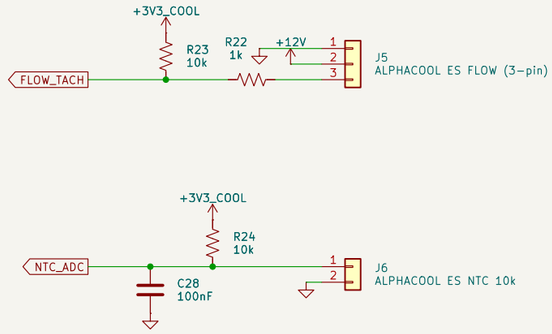
***Title: Signal Generator***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/19 |

Fan and pump: LEDC PWM at 25 kHz through a BSS138, because the PWM input can sit at 5.25 V (Intel 4-wire spec) and GPIOs are not 5 V tolerant. GPIO high = FET on = 0% drive; firmware inverts the duty.

Tach: open collector, 2 pulses/rev, counted with PCNT. Alphacool ES: tach -> GPIO18, NTC -> GPIO4.




**Cryocooler Controller**

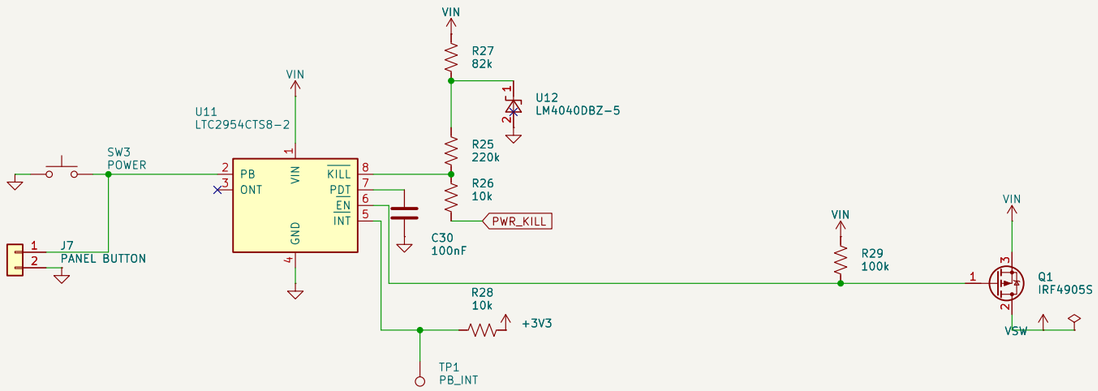
Sheet: /Water Cooling/
File: cooling.kicad_sch

***Title: Water Cooling***

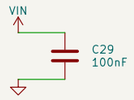
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/19 |

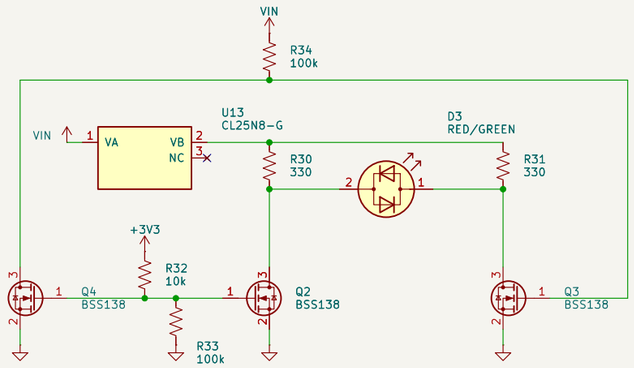
LTC2954-2: tap PB = on, hold = off. EN# drives the P-FET gate; KILL# from GPIO10 (active low, 100 ms pulse).

PWR_ON follows the +3V3 rail: on = green, standby = red. CL25 (25 mA) feeds both legs, ~9 mA reaches the LED, ~25 mA total from VIN in standby




LTC2954 DECOUPLING CAPACITORS

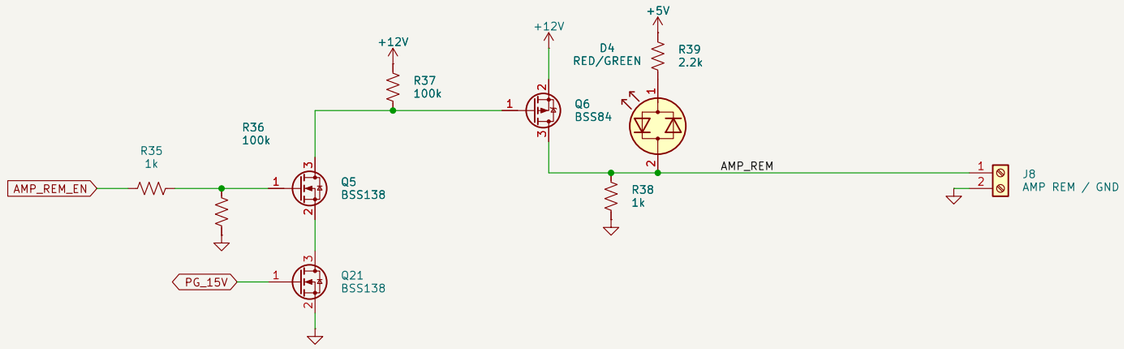



| | | |
|---|---|---|
| **Cryocooler Controller** | | |
| Sheet: /Power Button/ | | |
| File: power_button.kicad_sch | | |
| ***Title: Power Button*** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 5/19 |

GPIO6 high -> +12 V on the amplifier REM terminal (P-FET high-side switch), but only while PG_15V is high (both +-15 V rails good: Q9 in series with the level shifter). LED: green = on, red = off.

Amplifier: AUDIOZERONE ZE500.1 (external): REM terminal + RCA/line input. Its power leads do not pass through this board.







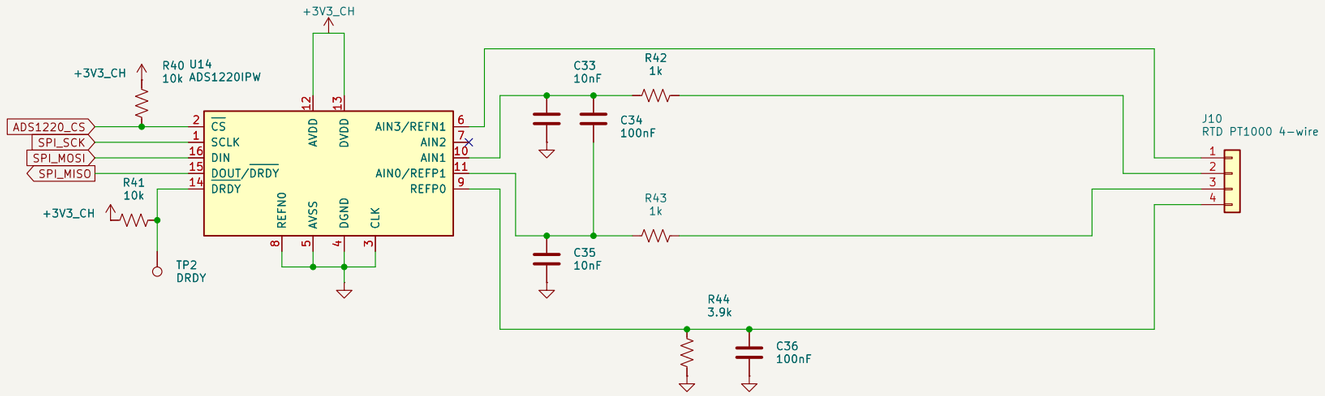
**Cryocooler Controller**

Sheet: /Amp Control/
File: amp_control.kicad_sch

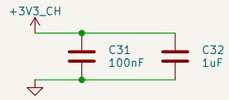
***Title: Amp Control***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 6/19 |

PT1000 4-wire ratiometric: IDAC1 (AIN3) excites, AIN1/AIN0 sense, Rref (3.9k) on REFP0/REFN0. SPI mode 1 on the main bus (CS GPIO44), internal oscillator (CLK tied low). DRDY is not wired to the MCU.



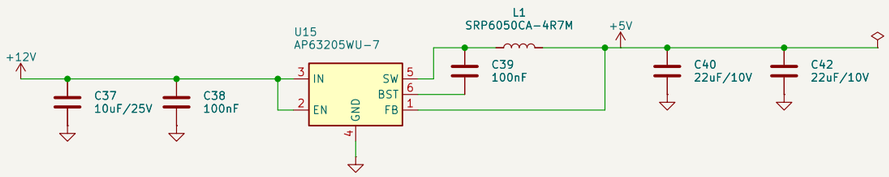

ADS1220 DECOUPLING CAPACITORS

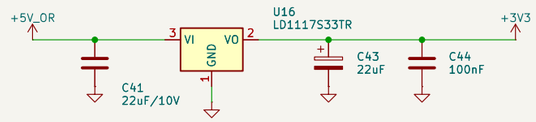



| | | |
|---|---|---|
| **Cryocooler Controller** | | |
| Sheet: /Cold Head Temp/<br>File: cold_head.kicad_sch | | |
| ***Title: Cold Head Temp*** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 7/19 |

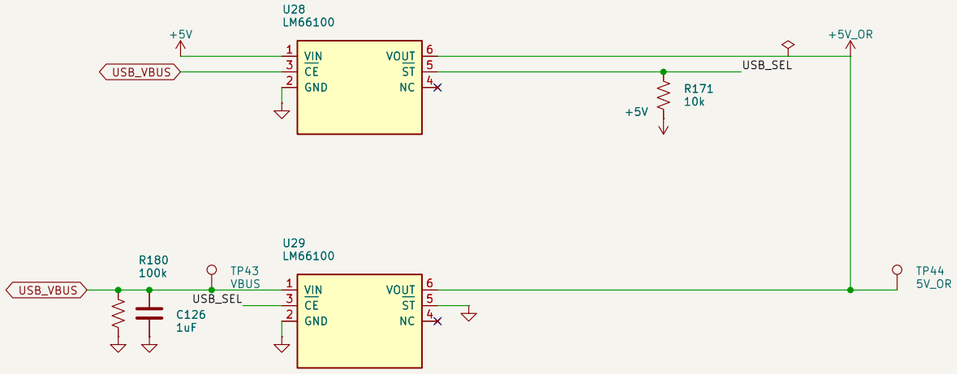
12 V -> 5 V (AP63205 buck, datasheet Fig. 1 / Table 3: 4.7 uH, 10 uF in, 2 x 22 uF out, 100 nF bootstrap) -> 3.3 V (LD1117S33).

AP63205: VIN 3.8–32 V (40 V for 400 ms), EN tied to VIN, FB tied to the fixed 5 V output.



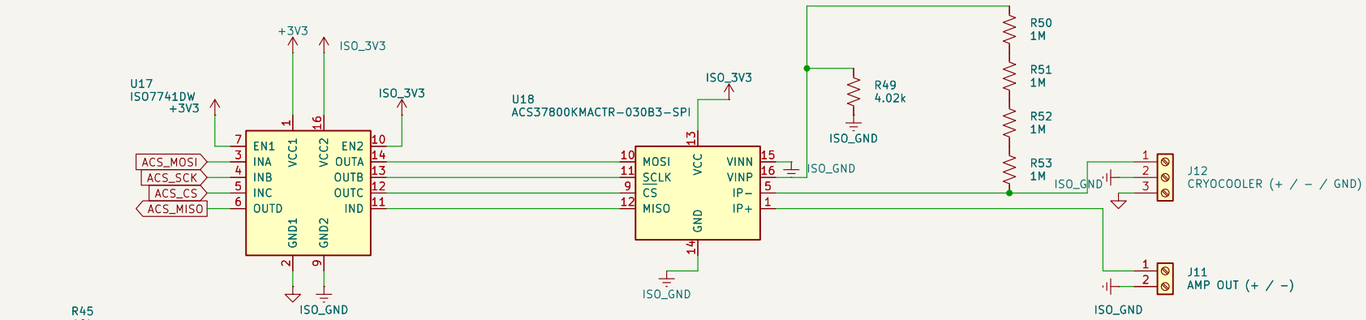

+5V_OR: USB alone powers +3V3 and the DDS, not the +5V rail or the +−15 V stage. LM66100 SLVSEZ8A Fig. 17 (CE to the other input, ST + pull-up to the second CE).

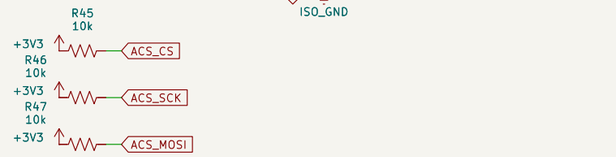


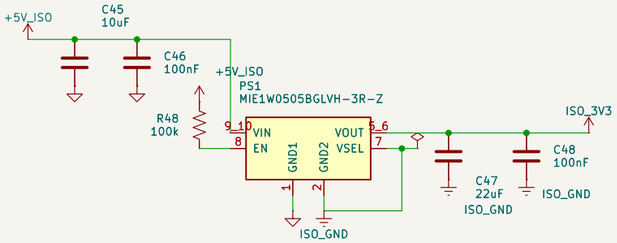

| | | |
|---|---|---|
| **Cryocooler Controller** | | |
| Sheet: /Rails 5V 3V3/ | | |
| File: rails_logic.kicad_sch | | |
| ***Title: Rails 5V 3V3*** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 8/19 |

Amplifier output V/I monitor: ACS37800 (SPI) on an isolated 3.3 V domain (MIE1W0505, VSEL=GND2 -> 3.3 V); SPI bus GPIO11/12/13, CS GPIO17.

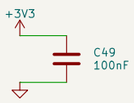
Mains-class creepage/clearance applies between ISO_GND domain and the logic side on the PCB.



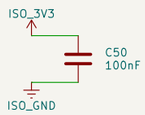



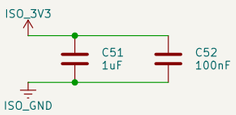

## ISO7741 VCC1 DECOUPLING CAPACITORS




## ISO7741 VCC2 DECOUPLING CAPACITORS




## ACS37800 DECOUPLING CAPACITORS




**Cryocooler Controller**

Sheet: /Amp Current/

File: amp_current.kicad_sch

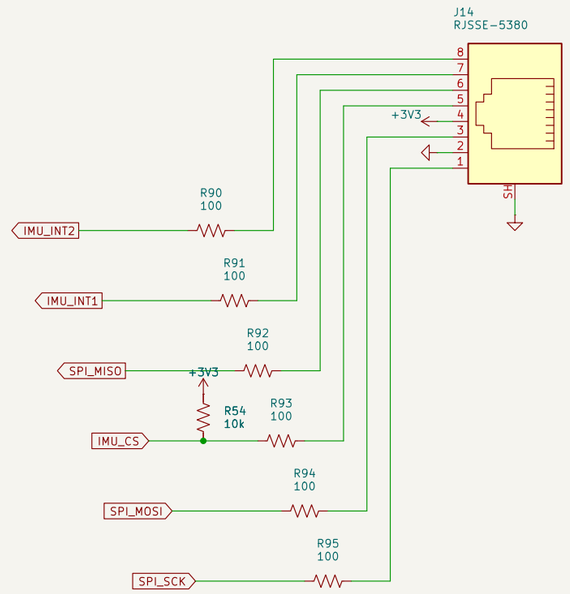
***Title: Amp Current***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 9/19 |

LSM6DSOX is the off-board Adafruit module (product 4438) on the cooler, cabled to J14 (RJ45, straight Cat5e cable) over the main SPI bus (CS GPIO1, INT1 GPIO2 single-tap, INT2 GPIO39 wake-up).

RJ45 pin -> module pad: 1 SCL, 2 GND, 3 SDA, 4 Vin, 5 CS, 6 DO, 7 I1, 8 I2 (3Vo pad open). Pairs: SCK with GND, MOSI with MISO, +3V3 with CS, INT1 with INT2. Plain jack, no magnetics.

Module Vin = +3V3 keeps its logic at 3.3 V. 100 R series resistors damp the cable; the 10 k keeps CS high with the cable unplugged; the jack shield is tied to GND.




**Cryocooler Controller**

Sheet: /IMU/
File: imu.kicad_sch

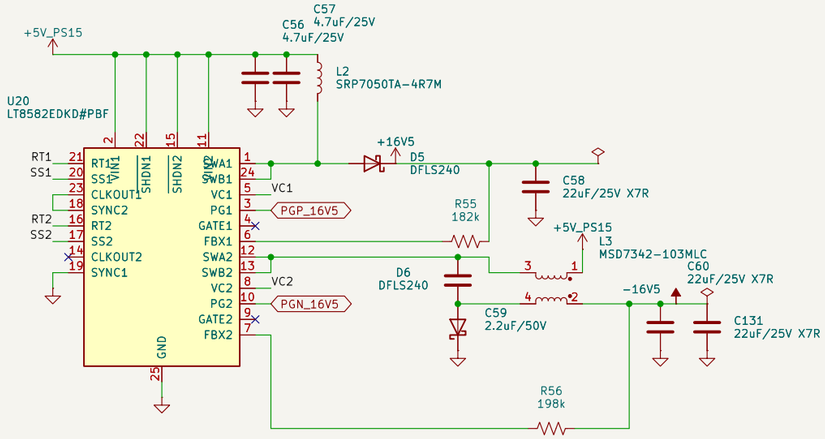
***Title: IMU***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 10/19 |

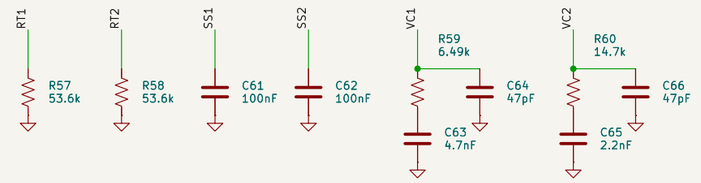
LT8582 (5 V in, 1.5 MHz): channel 1 boosts to +16.4 V, channel 2 inverts to −16.5 V; one LT3097 post-regulates them to +/−14.7 V.

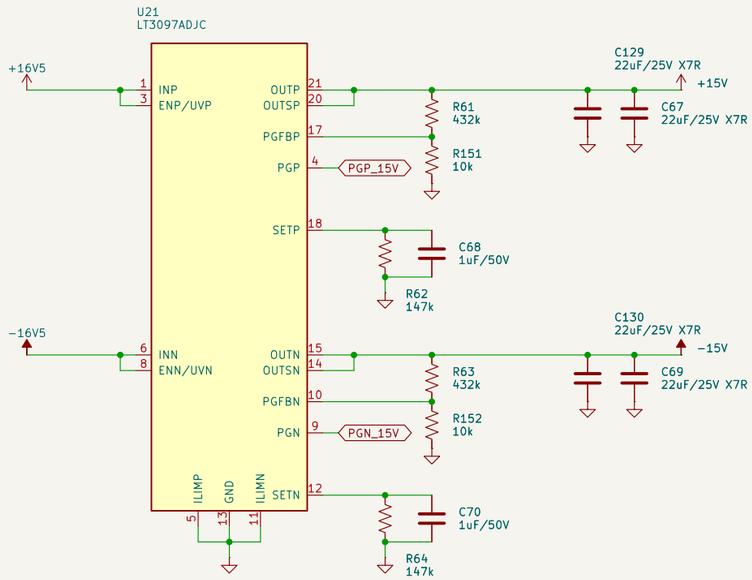
LT8582 circuit after datasheet p.30 (5 V to +/−12 V): RT 53.6k, SS 0.1 uF, VC 6.49k+4.7nF / 14.7k+2.2nF with 47 pF. RFBX1 = (16.4 − 1.204) / 83.3 uA = 182k, RFBX2 = (16.5 + 0.007) / 83.3 uA = 198k. Inverting needs L >= 5.6 uH (duty ~0.80): 10 uH coupled MSD7342.

PG1 / PG2 (open drain, high = within 4 %) and the LT3097 PGP / PGN go to the power_good sheet; GATE1/2 unused (no output disconnect).




## LT8582 TIMING AND COMPENSATION




**Cryocooler Controller**

Sheet: /Rails +−15V/
File: rails_15v.kicad_sch

***Title: Rails +−15V***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 11/19 |

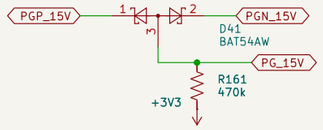
PGP and PGN pull their own node low when bad. Each node is the +15 V rail through 220k/62k (about 3.2 V when good, under 0.3 V if its pin pulls low or the rail is missing) and lights its green LED.

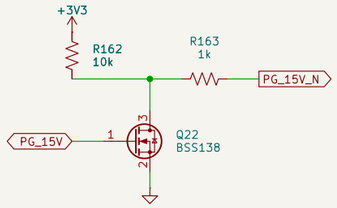
PG_15V = the two nodes ANDed by a BAT54AW with a 470k pull-up to +3V3 (weak so it cannot lift dead nodes; 3.3 V only when both are good); it gates the amplifier REM switch and reads LOW on GPIO43 (PG_15V_N) when both rails are good.

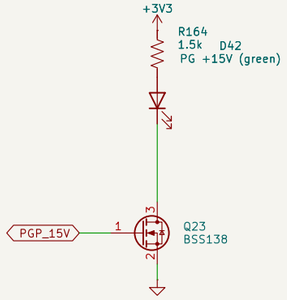


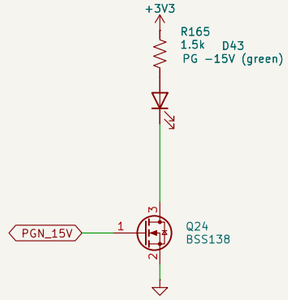




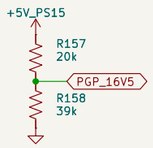


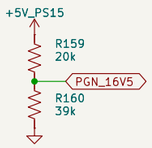




## LT8582 POWER GOOD (to MCP23S08 GP2 / GP3)



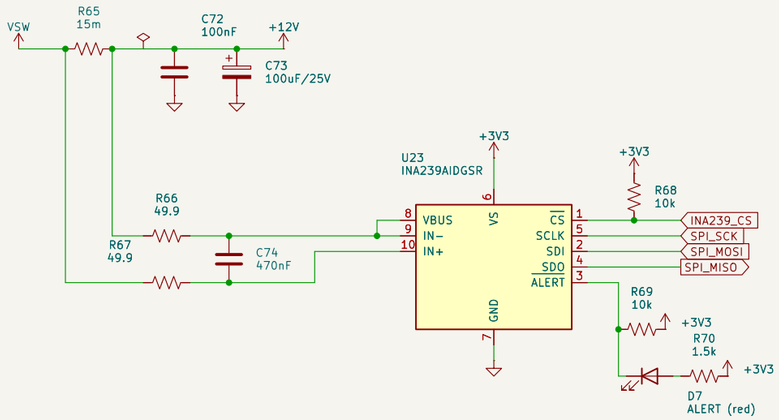

**Cryocooler Controller**

Sheet: /Power Good/
File: power_good.kicad_sch

***Title: Power Good***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 12/19 |

INA239 monitors the switched 12 V rail: 15 mOhm shunt, SPI mode 1 on the main bus (CS GPIO3), +/−163.84 mV range (~10.9 A). ALERT only drives its LED.




## 3V3 POWER LED




## INA239 DECOUPLING CAPACITORS




**Cryocooler Controller**

Sheet: /System Current/
File: system_current.kicad_sch

***Title: System Current***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 13/19 |

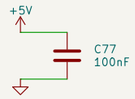
READY (MCP23S08 GP5) and FAULT (GP6) LEDs, active high; WS2812B status LED on GPIO38 via a 74AHCT1G125 3.3 V -> 5 V level shifter (WS2812B VIH = 0.7 x VDD).



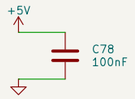


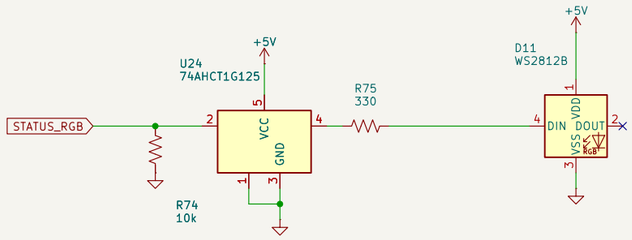


74AHCT1G125 DECOUPLING CAPACITORS




WS2812B DECOUPLING CAPACITORS



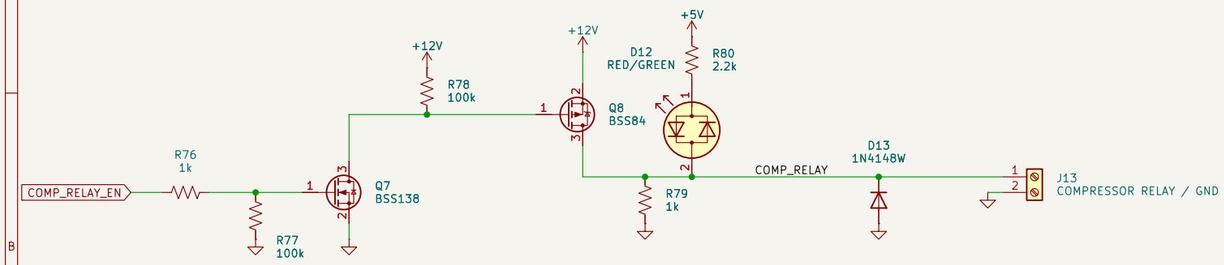

**Cryocooler Controller**

Sheet: /Indicators/
File: indicators.kicad_sch

***Title: Indicators***

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 14/19 |

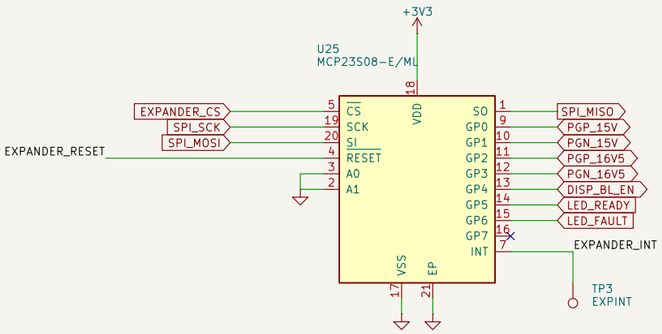
COMP_RELAY_EN (GPIO16) high -> +12 V on the compressor relay terminal (BSS84 high-side switch, 130 mA max: drive an external relay or SSR coil). Flyback diode across the output. LED: green = on, red = off.




| | | |
|---|---|---|
| **Cryocooler Controller** | | |
| Sheet: /Compressor Relay/<br>File: compressor_relay.kicad_sch | | |
| ***Title: Compressor Relay*** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 15/19 |

MCP23S08: SPI mode 0 slave on the main bus (CS GPIO8). SO releases within 100 ns of CS rising (datasheet Fig. 2–6, parameter 14), so it shares SPI_MISO without a buffer. A1/A0 low: opcodes 0x40 / 0x41.

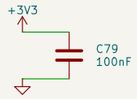
All pins power up as inputs. RESET, A1 and A0 must be biased (datasheet Table 1–1). Exposed pad to VSS, in addition to pin 17. GP7 is unused: firmware enables its weak pull-up.




## PULL-UPS




## MCP23S08 DECOUPLING CAPACITOR




**Cryocooler Controller**

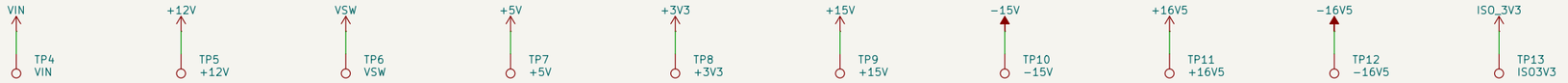
Sheet: /GPIO Expander/
File: gpio_expander.kicad_sch

***Title: GPIO Expander***

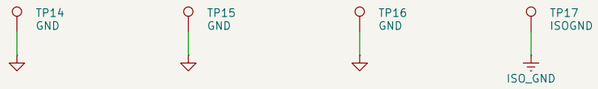
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 16/19 |

Debug aids: test points (pad per net, loops for ground) and rail LEDs. Each test point sits on a global label or power symbol; the PCB generator places it beside a pad on the same net.

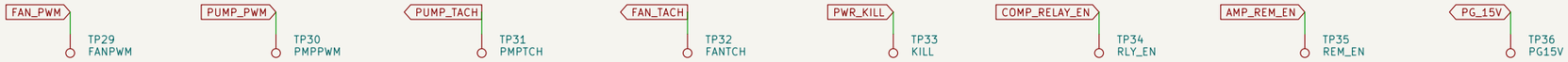
## POWER RAILS




## GROUND (scope loops)



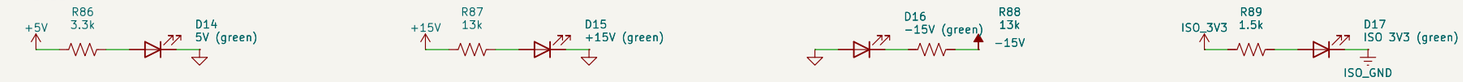

## FAN, PUMP, INTERRUPTS AND CONTROL




## ANALOG AND SENSORS




## RAIL LEDS




**Cryocooler Controller**

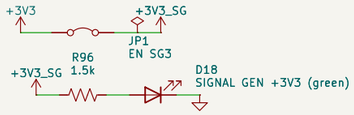
Sheet: /Debug/
File: debug.kicad_sch

***Title: Debug***

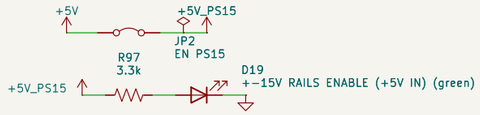
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 17/19 |

Each Keystone 5110 0603 jumper sits in series with one load supply: remove it to power that part off. The LED shows the switched rail is present. A part left unpowered can still be fed through its signal pins, so disconnect the cable too when you need it fully dead.

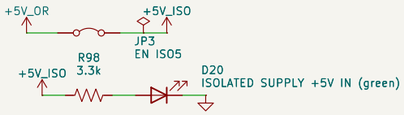
## SIGNAL GEN +3V3



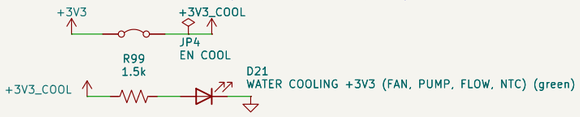

## +−15V RAILS ENABLE (+5V IN)



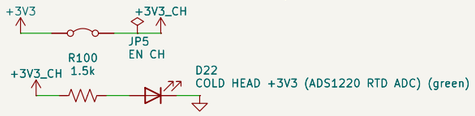

## ISOLATED SUPPLY +5V IN




## WATER COOLING +3V3 (FAN, PUMP, FLOW, NTC)




## COLD HEAD +3V3 (ADS1220 RTD ADC)




**Cryocooler Controller**

Sheet: /Supply Switches/
File: supply_switches.kicad_sch

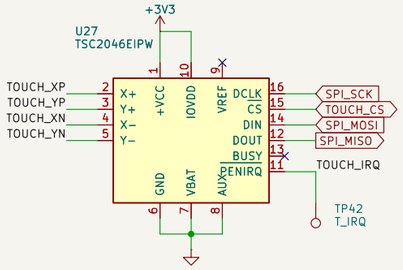
***Title: Supply Switches***

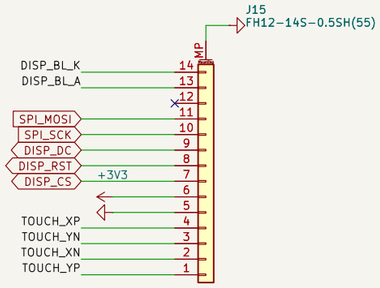
| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 18/19 |

Phoenix DCT T 2,4 QVGA S RTOUCH (in the UCS 125-87-F-GD-D2,4 lid), datasheet 109921: ST7789V, 4-wire SPI half duplex (SDA is input only), VCC 2.6-3.3 V, FPC FH12-14S-0.5SH.

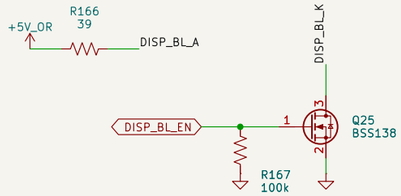
Pin 1 YU, 2 XL, 3 YD, 4 XR (raw resistive touch), 5 GND, 6 VCC, 7 /CS, 8 /RST, 9 RS (D/C), 10 SCL, 11 SDA, 12 NC, 13 A, 14 C. Backlight: 4 LEDs in parallel, 3.0 V / 60 mA typ.

GPIO45 (TOUCH_CS) and GPIO46 (DISP_RST) are strapping pins: 10k pull-downs hold their default (low) levels at reset; firmware drives TOUCH_CS high before any SPI traffic.

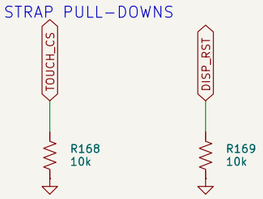


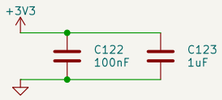

## BACKLIGHT



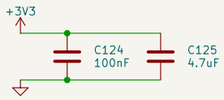

## STRAP PULL-DOWNS



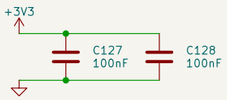

## TSC2046 DECOUPLING CAPACITORS




## DISPLAY DECOUPLING CAPACITORS




## SN74LV1T34 DECOUPLING CAPACITORS




## DISPLAY LOGIC BUFFERS

GPIO47/48 are 1.8 V on N32R16V. SN74LV1T34: VIH 1.36 V at 3.3 V.




| | | |
|---|---|---|
| Cryocooler Controller | | |
| Sheet: /Display/ | | |
| File: display.kicad_sch | | |
| **Title: Display** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.6 | | Id: 19/19 |